

CBERSECURITY PROJECT DOCUMENTATION

Title: - Simple Text Encryption & Decryption Tool Using Python

Project Name:- Text Encryption & Decryption Tool

Objective:

To develop a simple Python program that encrypts and decrypts text using Base64 encoding. This project demonstrates the basic concept of data security and information protection.

Problem Statement:

Sensitive information should not be shared in plain text because anyone can read it. This project converts normal text into encoded text and decodes it back when required.

Research Findings:

- 1. Encryption helps protect data from unauthorized access.*
- 2. Base64 encoding converts readable text into encoded format.*
- 3. Real cybersecurity uses stronger algorithms like AES and RSA for better security.*
- 4. This project is created for learning and demonstration purposes.*

Methodology:

- 1. User enters a text.*
- 2. User selects Encrypt or Decrypt.*
- 3. The program processes the text using Python's Base64 library.*
- 4 .The result is displayed on the screen.*

Tools Used:

- 1. Python 3*
- 2. Pydroid 3*

3. Base64 Library

Implementation:

- *Created a Python program.*
- *Used `base64.b64encode()` for encoding.*
- *Used `base64.b64decode()` for decoding.*
- *Displayed the output in the terminal.*

Screenshots:

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main.py*

/storage/emulated/...



```
1 import base64
2
3 print("=== Encryption & Decryption Tool ===")
4
5 text = input("Enter your text: ")
6
7 print("\n1. Encrypt")
8 print("2. Decrypt")
9
10 choice = input("Enter choice (1/2): ")
11
12 if choice == "1":
13     encrypted = base64.b64encode(text.
14     encode()).decode()
15     print("\nEncrypted Text:")
16     print(encrypted)
17 elif choice == "2":
18     try:
```

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```
=== Encryption & Decryption Tool ===  
Enter your text: aGVsbG8gU25laGE=
```

```
1. Encrypt
```

```
2. Decrypt
```

```
Enter choice (1/2): 2
```

```
Decrypted Text:
```

```
hello Sneha
```

```
[Program finished]
```

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```
=== Encryption & Decryption Tool ===  
Enter your text: hello sneha
```

```
1. Encrypt  
2. Decrypt  
Enter choice (1/2): 1
```

```
Encrypted Text:  
aGVsbG8gc25laGE=
```

```
[Program finished]
```

Results:

The program successfully encrypts plain text into encoded text and decrypts it back to the original text.

Challenges:

- *Understanding Base64 encoding.*
- *Handling invalid input during decryption.*
- *Running and testing the project on a mobile device.*

Future Scope:

- *Add password protection.*
- *Support file encryption and decryption.*
- *Develop a graphical user interface (GUI).*
- *Implement advanced encryption algorithms such as AES.*

Conclusion:

This project successfully demonstrates the basic concept of text encryption and decryption using Python. It helps beginners understand how data can be protected before being shared.